

Swish Analytics NFL Sack Probability Prediction Challenge

Analysis Memo by Kevin Le

Project Overview & Objective

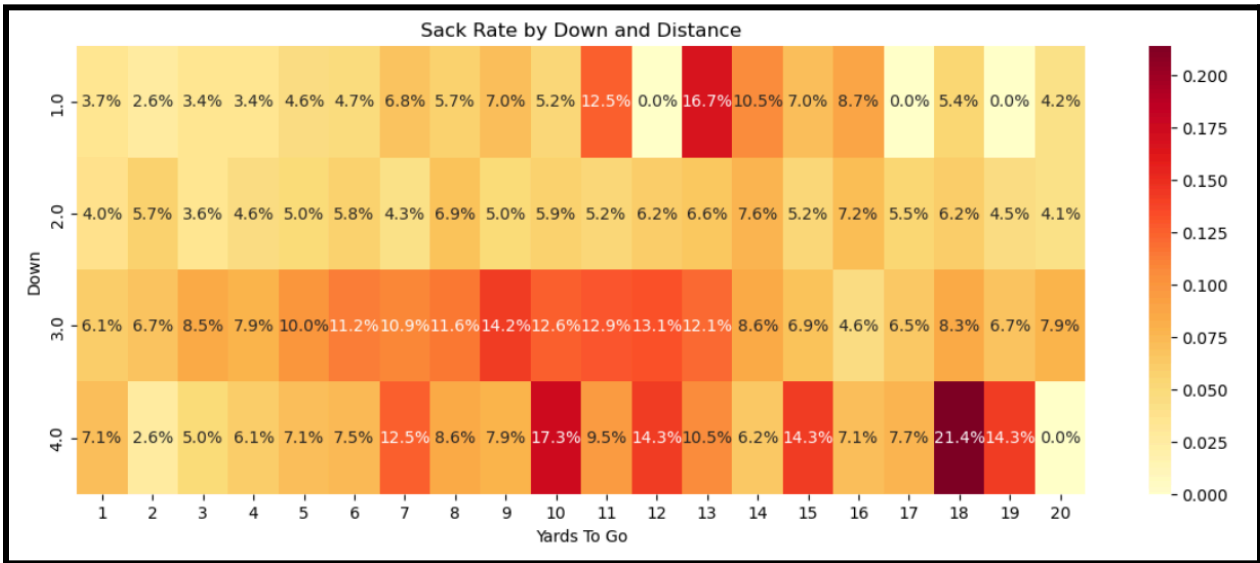
The goal of this project was to analyze play-by-play (PBP) data and player metadata from the 2021–2023 NFL seasons to build a predictive model estimating the probability that a given pass play results in a sack. This framework routes raw sports data through a pipeline of cleaning, exploratory analysis, and predictive modeling to surface actionable, situational insights.

Data Cleaning & Integration (Summary)

The data pipeline successfully resolved identity mismatches across play-by-play and roster tables by leveraging GSIS player IDs as relational bridges. Non-relevant events (spikes, runs, special teams) were filtered out, strictly isolating active dropbacks (`play_type == 'pass'`) to establish a clean, accurate modeling subset. Calculated fields for rolling sack rate (last 120 dropbacks) was created for passers, offenses, and defenses as a catch all predictive feature.

Key Takeaways from Exploratory Data Analysis (EDA)

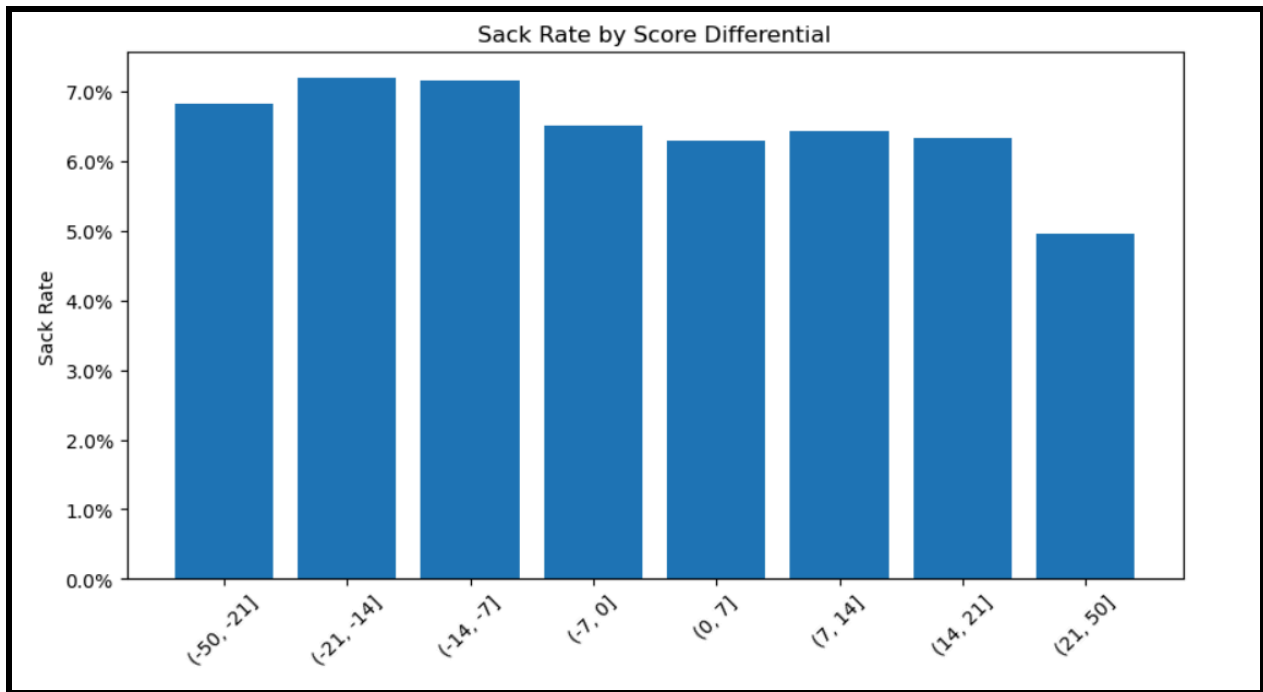
- **Situational & Contextual Drivers:**
 - **Down and Yards to Go:** These emerged as incredibly strong predictors of sack probability. Passing on obvious, long-yardage downs heavily favors the pass rush, as defensive schemes can pin their ears back without honoring the run. The heatmap below also shows potential interaction effects between the two, which could indicate that tree based models with outperform the logistic regression.



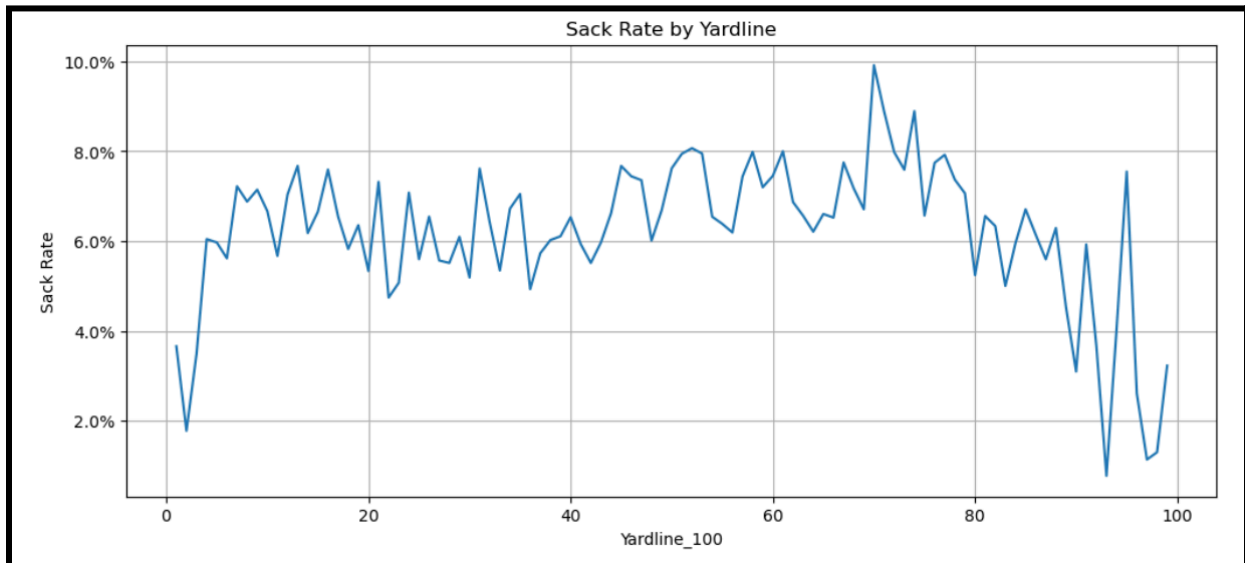
- **Score Differential:** Game state heavily influences sack rates. Teams trailing big in the game are forced into predictable, deep-passing situations, significantly elevating their sack exposure. Teams up big are likely running the ball or calling safe plays, leading to lower sack rates.

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- **Yardline:** Field position dictates play calling and thus sack rate. Sack rates are relatively stable between the twenties but drop dramatically near each of the goal lines, likely because offenses are more likely to call a run or call a short yardage pass play that does not give the pass rush time to get to the quarterback.

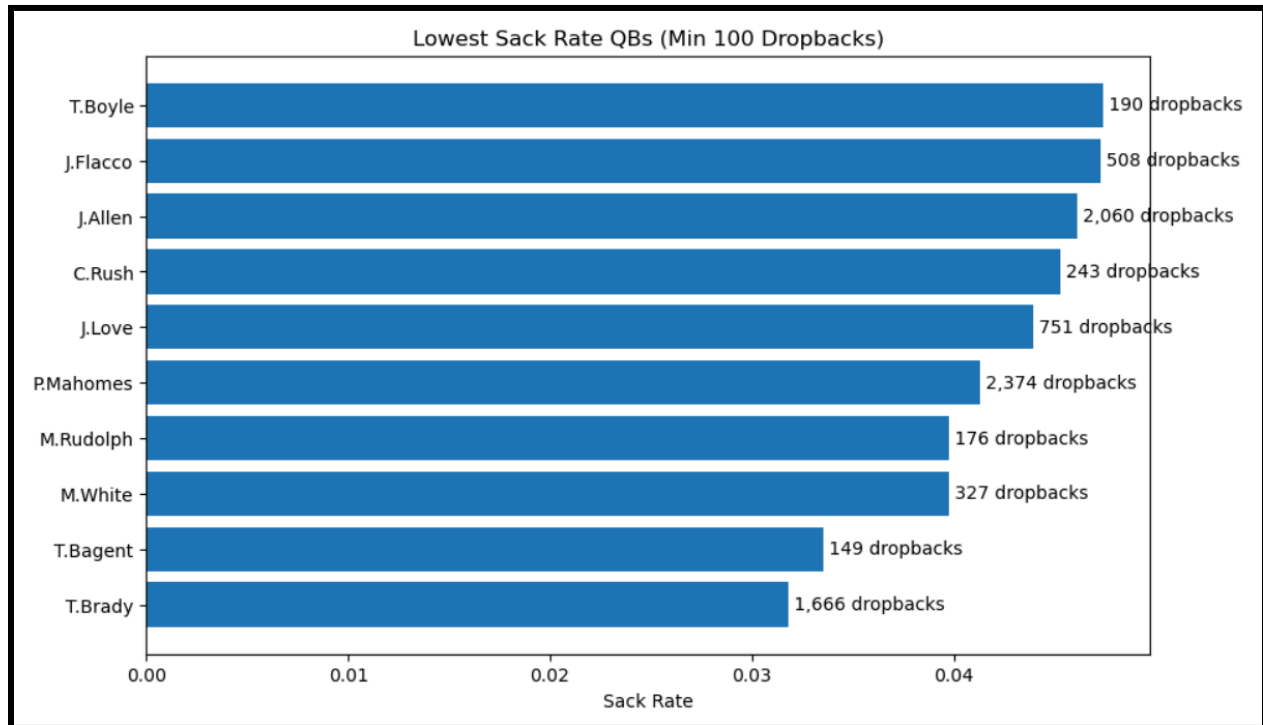


- **Quarterback Quality & Systemic Profiles:**

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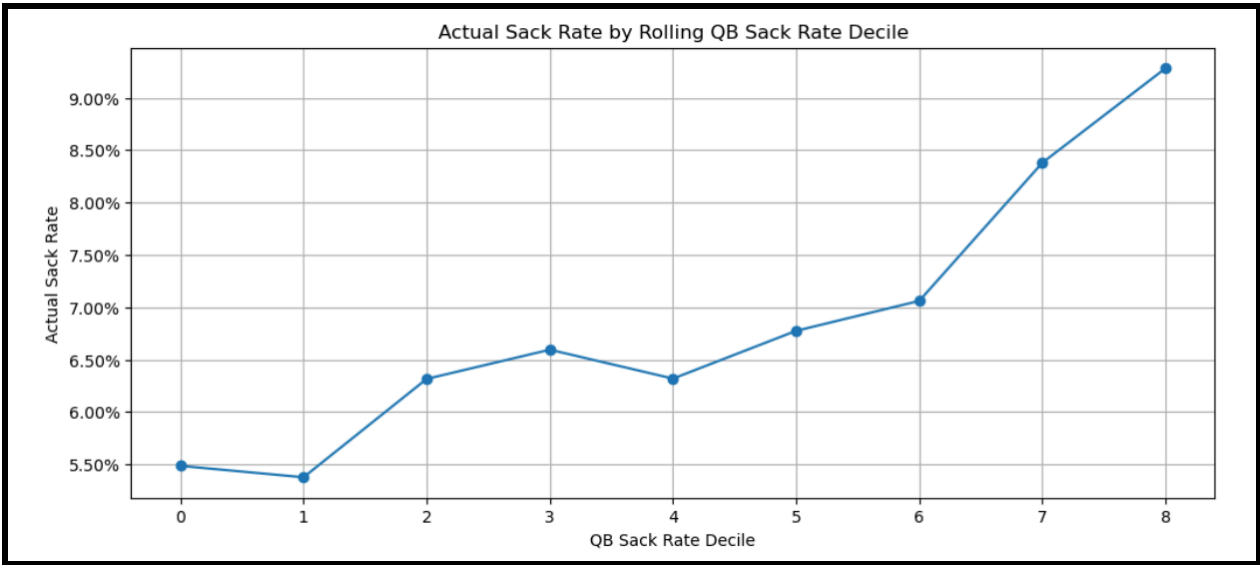
- Evaluating the **top 10 and bottom 10 sack rates** highlighted that sacks are highly quarterback-dependent traits. Elite pocket-presence signal-callers consistently mitigate pressure and maintain low sack profiles, whereas journeyman and lower tier quarterbacks dominate the highest sack rate tiers, likely because they fail to diagnose pre-snap coverage.



- Specific quarterback names were not included in the model to avoid overfitting to specific quarterbacks, but in future models a metric to calculate QB quality could be a helpful predictor.
- **Dynamic Performance Baselines:**
 - Calculating rolling averages over the **last 120 pass plays** successfully captured shifting momentum and systemic form, smoothing out single-play variance to isolate true baseline vulnerability for the QB, offense, and opposing defense.
 - In analyzing the calculated fields, the deciles chart below sorts all plays from lowest to highest rolling QB sack rate and splits them into 10 equally-sized groups. Then for each decile we calculate the sack rate. As we can see, as the rolling sack rate increases, so does the actual sack rate which means this is a pretty good potential predictor.

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Model Performance & Evaluation Insights

The modeling architecture relied heavily on chronological validation splits to prevent data leakage. Given the high class imbalance of sacks, the models were evaluated primarily using the **Receiver Operating Characteristic - Area Under the Curve (ROC-AUC)**. The ROC AUC score represents the probability that a model will rank a randomly chosen positive example higher than a randomly chosen negative example.

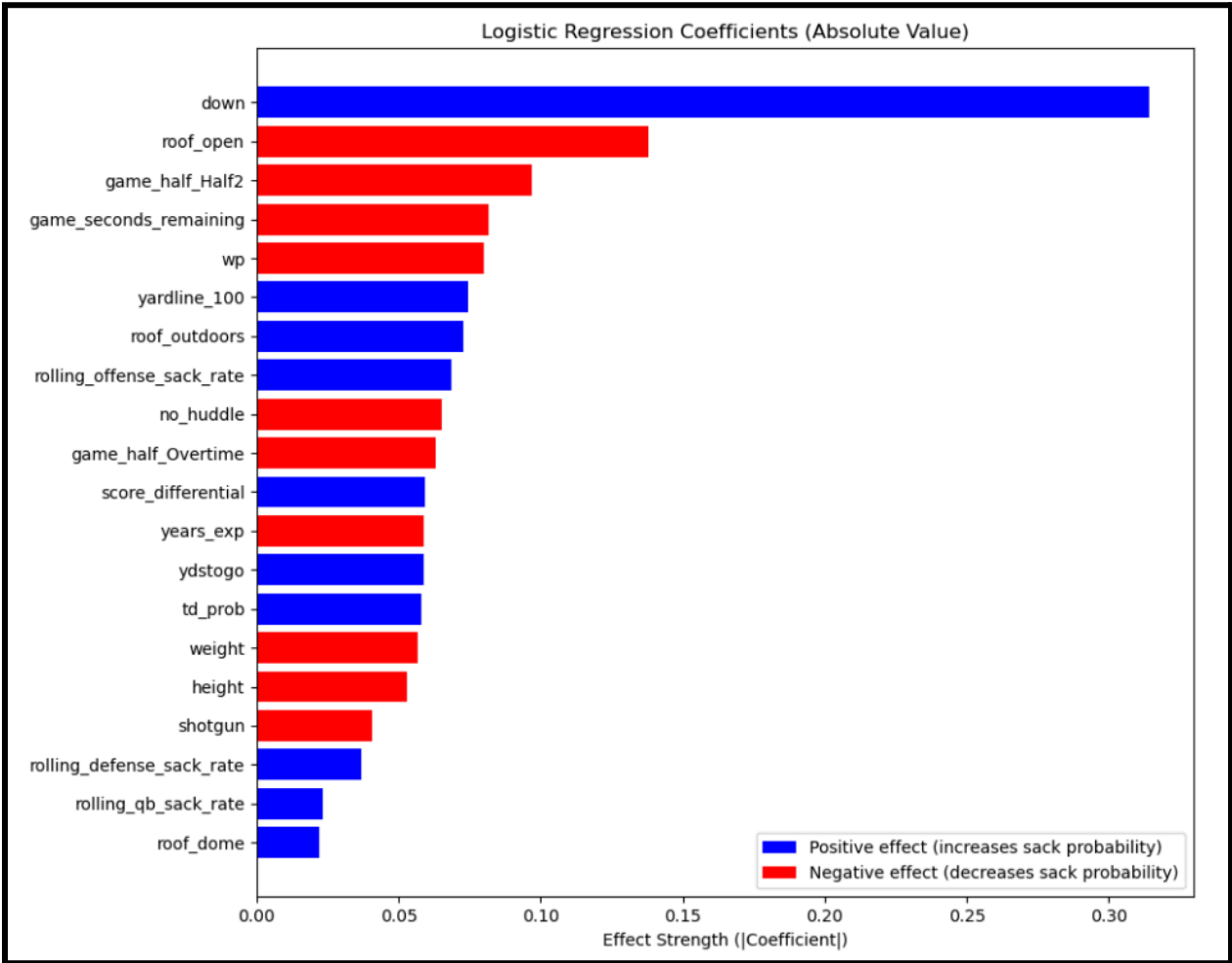
The secondary metric was **Log loss**, which evaluates how closely the predicted probabilities align with the actual binary labels, heavily penalizing confident but incorrect predictions.

Model Type	Key Prioritized Features	Core Model Performance & Takeaways
Logistic Regression	Down was the feature with the strongest coefficient (positive coefficient). Roof open, game seconds remaining, and win probability all had negative coefficients and were the next strongest features. The next three strongest features were all positive: yardline_100, roof outdoors, and rolling	• ROC-AUC: 0.6071 • Log loss: 0.2502 As a baseline model, an ROC above 0.5 means that we are providing value in this model above just guessing blindly.

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	offense_sack_rate (calculated field).	The Log loss is also slightly better than what would have been calculated had we predicted the league average sack rate for every play (0.255).
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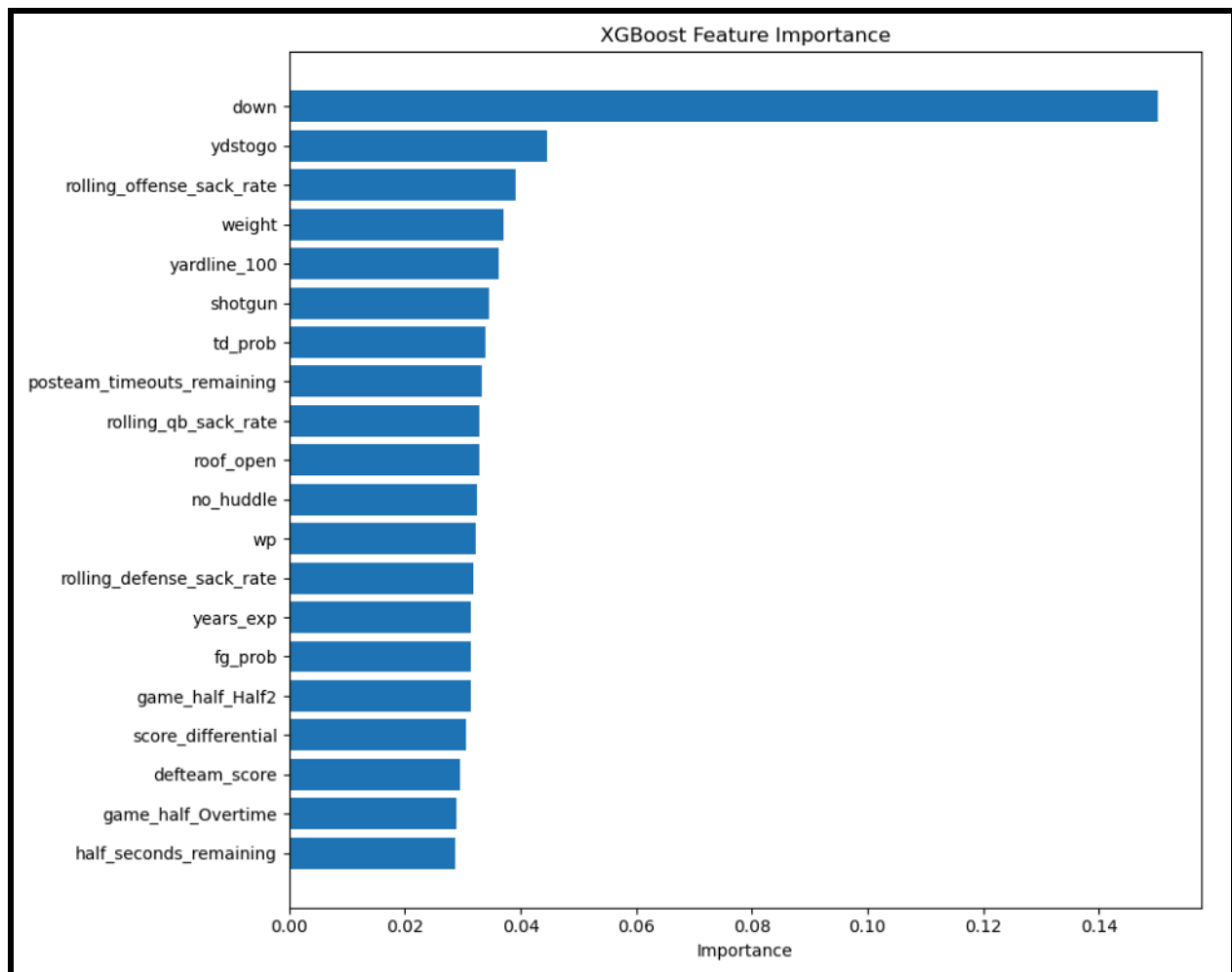
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Model Type	Key Prioritized Features	Core Model Performance & Takeaways
XGBoost Classifier	Down was the by far feature with the strongest importance. The next most important features were yards to go, rolling offense sack rate (calculated field), weight, yardline_100, and shotgun formation. These features align more with the expected predictors of sack rate and mostly track to what was revealed in the logistic regression as well.	• ROC-AUC: 0.6235 • Log loss: 0.2483 The metrics here were expected to be better than the logistic regression because tree based models account for nonlinear relationships and heavy interaction effects better. However, the model was only a little better, indicating the features primarily contain weak, additive relationships rather than strong nonlinear interactions.

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Future Enhancements

To maximize the predictive boundary of the model, access to granular tracking and proprietary datasets would target the following critical blind spots:

- **Pre-Snap Pass Rusher and Pass Blocker Alignment:** Integrating tracking data to pinpoint exactly who is aligned on the line of scrimmage, allowing the model to weight the pass rush and offensive line based on active personnel groupings rather than generic team metrics. I would also need to be able to merge this to pass rusher and pass blocker grades (next bullet).
- **Player Quality Tiers (PFF Grades):** Incorporating individual pass-blocking/pass-rushing and quarterback grades to establish a talent-baseline matchup matrix for every play.
- **Pre-Snap Numbers Advantage:** Quantifying the exact difference between the number of designated blockers and potential rushers showing blitz at the line of

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scrimmage—offering an immediate math-based indicator of impending pressure before the snap.

Contact

Find the full project [HERE](#) - the readme serves as a full report with visualizations.

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